Q2 工业电缆的选择及截面确定
Selection and sizing of industrial cables

先选择适合工业使用条件的电缆结构，再按载流量、压降及功率损耗三项准则确定导体截面。

Select cable construction for industrial service conditions, then size the conductor against current-carrying capacity, voltage drop and power loss.

【1】三类表格的用途 / Purpose of the tables

表格 / Table	用途 / Use
Appendix 7 **安装方式 / Installation methods**	按实际敷设方式及图示确定 Reference Method，供载流量表选栏；安装项目编号不一定等于参考方法编号。 Identifies the Reference Method from the actual installation and illustrated arrangement for selecting the current-rating column. Installation item numbers are not necessarily reference-method numbers.
Table A6(1) **载流量及压降 / Current ratings and voltage drop**	适用于有或无护套的单芯 PVC 绝缘、非铠装铜电缆。载流量页按截面、参考方法、回路及排列选值，基准为环境 30℃、导体 70℃；压降页按截面及相应回路／排列查 mV/(A·m)值。 Applies to single-core PVC-insulated, non-armoured copper cables, with or without sheath. The current-rating page uses size, Reference Method, circuit and formation, assuming 30℃ ambient and 70℃ conductor temperature. The voltage-drop page gives mV/(A·m) by size and applicable circuit/formation.
Tables A5(1)–A5(6) **修正系数 / Correction factors**	为载流量核算及保护配合提供修正系数：A5(1)/(2)环境温度、(3)成组、(4)隔热包覆、(5)保护器件、(6)封闭地沟；各表按适用条件选用。 Provide factors for current-capacity assessment and protective-device coordination: A5(1)/(2) ambient temperature, (3) grouping, (4) thermal insulation, (5) protective devices and (6) enclosed trenches. Apply each within its scope.

【2】电缆选型因素 / Factors governing cable selection

因素 / Factor	定性解释 / Qualitative explanation
导体、芯数及绝缘 **Conductor, cores and insulation**	按用途选择铜／铝、芯数及 PVC／XLPE 绝缘；电阻、允许温度及额定数据随结构改变，载流量和压降资料必须匹配。 Select conductor material, core count and PVC/XLPE insulation for the service. Resistance, permissible temperature and ratings vary; use matching current and voltage-drop data.
外部影响、护套及铠装 **External influences, sheath and armour**	工业场所的潮湿、油污、化学品、粉尘及机械损伤风险，决定护套、铠装和防护方式，确保电缆适合现场使用。 Industrial moisture, oil, chemicals, dust and mechanical-damage risks govern sheath, armour and protection, ensuring suitability for the actual environment.
安装方式及排列 **Installation method and arrangement**	敷设方式改变散热条件。按 Appendix 7 确定参考方法，并匹配相制、载流导体数量及排列。例如，A6(1)的三相 4mm² 单芯 PVC 铜缆，Method 3 表面管道为 28A，Method 4 隔热墙内管道为 24A。 Installation affects cooling. Use Appendix 7 and match phase, loaded conductors and formation. For example, A6(1) rates 4mm² single-core PVC copper cables in a three-phase circuit at 28A for Method 3 surface conduit and 24A for Method 4 conduit in an insulated wall.

【3】导体截面确定因素 / Factors governing conductor sizing

因素 / Factor	定性解释 / Qualitative explanation
负载及设计电流 Ib **Loading and design current**	按设备负载及实际使用情况确定 Ib；载流能力须满足负载，负载增大通常需增大截面，以限制导体温升。 Determine Ib from equipment loading and service conditions. Capacity must meet the load; higher current generally requires a larger cross-section to limit temperature rise.
环境温度 Ca **Ambient temperature**	温度升高使可用载流量降低。一般用 A5(1)；BS3036 半封闭熔断器承担过载保护时用 A5(2)，两表二选一。例：70℃ PVC 电缆、45℃、MCB 20A 且其他系数为 1 时，Ca=0.79，所需 It=20/0.79=25.3A。 Higher temperature reduces capacity. Use A5(1) generally, or A5(2) for overload protection by a BS3036 semi-enclosed fuse. For example, for 70℃ PVC cables at 45℃ with a 20A MCB and other factors 1, Ca = 0.79 and required It = 20/0.79 = 25.3A.
电缆成组 Cg **Grouping**	成束或围护内回路互相加热。按 A5(3)的回路数量、间距及排列选 Cg；散热越受限，通常越需增大截面。 Bunched or enclosed circuits heat one another. Select Cg from A5(3) for circuit count, spacing and arrangement; restricted cooling generally requires a larger size.
隔热材料 Ci **Thermal insulation**	隔热材料阻碍散热。按实际安装条件及 A5(4)的适用范围折减；参考 Method 已计入的同一隔热影响，应避免重复修正。 Thermal insulation restricts cooling. Use the actual installation and applicable A5(4) conditions, avoiding duplicate correction for an effect already included in the reference Method.
保护器件额定值 In 及 Cd **Overcurrent protective device**	按 In 及适用系数求所需表列载流量，使电缆与保护配合。A5(5)的 Cd：BS3036 半封闭熔断器为 0.725，其他所列器件为 1.0。 Coordinate cable capacity with In and applicable factors. A5(5) gives Cd = 0.725 for BS3036 semi-enclosed fuses and 1.0 for the other listed devices.
封闭地沟 **Enclosed trenches**	封闭地沟积热影响散热。按沟体、盖板、导体截面、电缆数量及间距选 A5(6)系数，修正相应 Method 12／13 的基础额定值。 Enclosed trenches retain heat. Apply A5(6) for trench construction, cover, conductor size, cable count and spacing to the appropriate Method 12/13 base rating.
长度及压降 **Cable length and voltage drop**	长度和电流越大，压降通常越大。按匹配数据核算，并计入上游压降；载流量合格后仍可能需增大截面。例：A6(1)的三相 4mm² Method 3 电缆，压降系数为 9.5mV/(A·m)，20A、40m 时压降=9.5×20×40/1000=7.6V。 Longer routes and higher current increase voltage drop. Use matching data and include upstream drop; adequate current capacity may still require a larger size. For example, A6(1) gives 9.5mV/(A·m) for a 4mm² three-phase Method 3 cable: at 20A over 40m, drop = 9.5×20×40/1000 = 7.6V.
功率损耗及能源效益 **Power loss and energy efficiency**	工业回路长期带载时须考虑 I^2R 损耗。相同电流下，较大截面降低电阻和损耗，改善能源效益。 For long-running industrial circuits, consider I^2R loss. At a given current, a larger cross-section reduces resistance and losses, improving energy efficiency.

$$\mathbf{It = In / (Ca \times Cg \times Ci \times Cd)}$$

过载保护配合时，It 为所需最低表列载流量，选择 Ita ≥ It；按现场条件采用适用系数。

For overload coordination, It is the minimum required tabulated capacity; select Ita ≥ It using the applicable factors.